

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 Total Marks:50

Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I A. Govardhan Reddy (192421359) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Govardhan Reddy
Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message A cybersecurity intern intercepts the coded message: “GSRH RH Z HVXIVG” during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

Python helper functions (Question 2)

```
import string; A = string.ascii_uppercase
def caesar(t,k): return ''.join(A[(A.index(c)+k)%26] if c in A else c for c in t)
def atbash(t): return ''.join(A[25-A.index(c)] if c in A else c for c in t)
```

Question 2: Intercepted Suspicious Message

Scenario: The intercepted message is “GSRH RH Z HVXIVG”, which suggests a simple substitution. (a) Deduce the most likely original message. (b) Justify using repeating patterns, letter structures or known cipher characteristics.

Concepts integrated

Monoalphabetic substitution • Atbash (special affine cipher with $a = -1$, $b = 25$) • cryptanalysis by pattern words, letter frequency and elimination of Caesar • self-inverse ciphers.

a. Deduction of the plaintext

Method. (1) Observe word shape 4-2-1-6. (2) Test the Caesar hypothesis (constant shift). (3) Test the Atbash hypothesis, $P + C = 25$. (4) Decode and check that the result is meaningful English.

Atbash table (A<->Z, B<->Y ...): shaded letters are those used in the intercepted text

Cipher	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Plain	Z	Y	X	W	V	U	T	S	R	Q	P	O	N	M	L	K	J	I	H	G	F	E	D	C	B	A

Figure 1: Atbash substitution table

Cipher	Letter-by-letter (Atbash: $C = 25 - P$)	Plain
GSRH	G(6)→T(19), S(18)→H(7), R(17)→I(8), H(7)→S(18)	THIS
RH	R→I, H→S	IS
Z	Z(25)→A(0)	A
HVXIVG	H→S, V→E, X→C, I→R, V→E, G→T	SECRET

Original message: THIS IS A SECRET

b. Justification

Evidence	Observation	Conclusion
One-letter word	Z stands alone; English one-letter words are A or I	Z = A (opposite ends of alphabet → reversed alphabet)
Repeated word	“RH” appears twice (as word 2, and as the ending of GSRH)	Common word IS; GSRH → THIS also ends in IS
Repeated letters	V occurs twice inside HVXIVG (positions 2 and 5)	Pattern S-E-C-R-E-T: E repeats, same-letter→same-letter
Consistent mapping	H→S in GSRH and HVXIVG; R→I in GSRH and RH	Single fixed alphabet = monoalphabetic
Caesar test	Shifts G→T = +13, S→H = -11, Z→A = +1 (not constant)	Caesar rejected
Sum test	6+19, 18+7, 17+8, 7+18, 25+0 all = 25	Atbash confirmed

Tool implementation and output

```
>>> atbash('GSRH RH Z HVXIVG')
>>> [caesar('GSRH RH Z HVXIVG',k) for k in range(1,26)] # brute force
'THIS IS A SECRET'
k= 1 HTSI SI A IWYJWH k= 4 KWVL VL D LZBMZK k=13 TFEU EU M UIKVIT ... (none readable for k = 1..25)
```

Analysis & validation

- Atbash is its own inverse ($\text{atbash}(\text{atbash}(x)) = x$), so re-encoding the answer reproduces the intercepted text – a full validation.

- All 25 Caesar shifts were tested and none produced English, so the elimination argument is complete, not just a lucky guess.
- Atbash has no key, so anyone who guesses the algorithm reads the message – it gives encoding, not confidentiality.

Question 4: Suspicious Digital Asset Investigation

Scenario: A company logo file may contain hidden confidential information. (a) Analyze how investigators could detect hidden data. (b) Recommend two techniques to strengthen such hidden communications against detection or extraction.

Concepts integrated

Steganography (hiding existence) vs cryptography (hiding meaning) • spatial-domain LSB embedding and transform-domain (DCT/DWT) embedding • steganalysis (visual, structural, statistical) • digital-forensic integrity (hashing, chain of custody) • encryption + keyed pseudo-random embedding.

a. Detection methodology

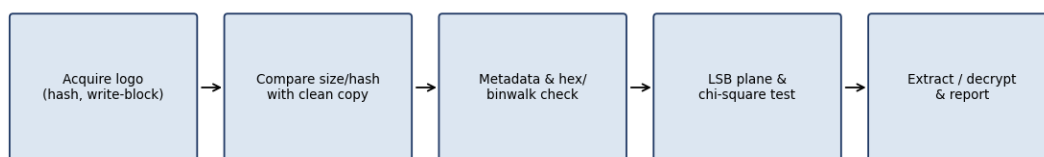


Figure 2: Systematic forensic workflow for the logo file

Stage	Technique / tool	What indicates hidden data
1 Preserve	sha256sum, write-blocker, work on a copy	Hash recorded for chain of custody
2 Baseline compare	Compare size/hash/pixels with the official logo	Larger size, different hash, scattered ± 1 pixel differences
3 Structure & metadata	ExifTool, xxd (hex), strings, binwalk	Data after PNG IEND / JPEG FFD9, odd chunks, embedded archives, suspicious comments
4 Visual / bit-plane	StegSolve, zsteg (LSB planes, channels)	Structured or text-like pattern in LSB plane, readable ASCII
5 Statistical	Chi-square pairs-of-values, RS / sample-pair, StegExpose	Histogram pair equalization or abnormal LSB statistics
6 Extract & report	steghide / OpenStego / dictionary attack, document findings	Recovered payload, hashes and screenshots in report

Tool implementation: controlled experiment (Python)

A 256×256 grayscale “logo-like” cover image was generated and a 1,800-byte message was hidden in the least-significant bits of its first 14,400 pixels (22 % capacity). The image was then analysed as an investigator would.

```

stego = (cover_flat & 0xFE) | payload_bits # LSB embedding
chi2 = sum((h[2i]-e)**2/e for pairs) ; e=(h[2i]+h[2i+1])/2 # pairs-of-values test
lsb_mean = (img[:20] & 1).mean()
  
```

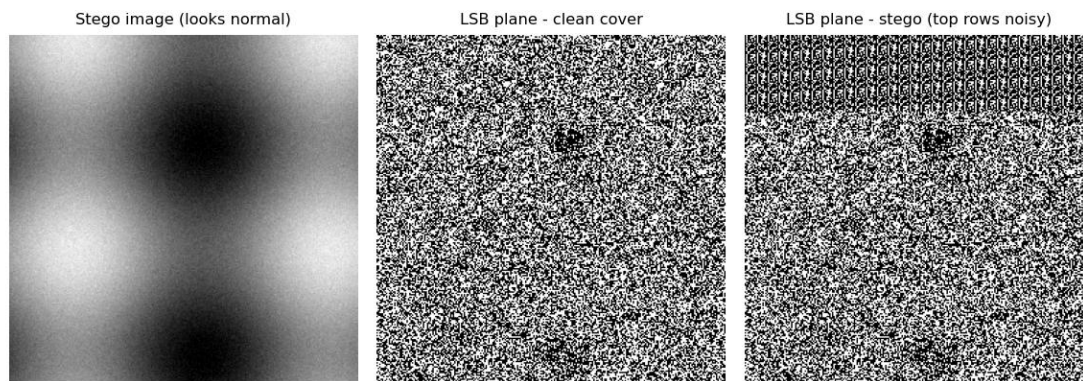


Figure 3: Stego image looks normal; LSB-plane analysis exposes the embedded rows

Image variant (first 20 % of pixels)	Pixels changed	Max diff	Chi-square	LSB mean	Detected?
Clean cover	0	0	41.9	0.506	No (baseline)
Plain-text payload, sequential LSB	7,205	1	474.0	0.371	Yes – strongly
Encrypted (random) payload, sequential	7,258	1	41.2	0.501	No – looks clean
Encrypted payload, key-scattered positions	7,204	1	46.9	0.501	No – looks clean

Extracting the first 30 bytes of the LSB stream from the plain-text variant returned “CONFIDENTIAL: Q4 LAUNCH 12 NOV”, proving extraction is trivial when the payload is unprotected.

Analysis & validation of detection

- The embedding changes pixels by at most 1 grey level, so it is invisible to the eye (Figure 3, left) but visible to a hash comparison and pixel-difference test.
- ASCII text has many zero bits, so the LSB mean drops from 0.506 to 0.371, and the chi-square statistic rises more than ten-fold (474 vs 42): both are strong indicators.
- Limit of the experiment: a synthetic image and a simple chi-square test; real logos (flat colours, anti-aliased edges) need RS / sample-pair analysis and comparison with the original, but the method is the same.

b. Two techniques to strengthen the hidden communication

- **1. Encrypt (and compress) the payload before embedding, e.g. AES-256-GCM.** Encrypted bits are statistically random, so the LSB mean (0.501) and chi-square (41.2) become indistinguishable from the clean cover in the experiment above, and even a successful extraction yields only ciphertext without the key.
- **2. Key-based pseudo-random, low-rate and adaptive/transform-domain embedding.** A secret stego-key seeds a PRNG that chooses scattered positions (sequential LSB is easy to find; the scattered variant also removed the localized anomaly: chi-square 46.9, no rows stand out). Embedding in textured regions or in DCT/DWT coefficients (spread spectrum) at a low rate resists RS/sample-pair analysis and survives JPEG compression or resizing.

Combined effect: confidentiality (encryption) + undetectability and key-dependent extraction (keyed scattered embedding) = defence in depth.

Summary of final answers

Q	Topic	Final answer
2	Atbash cipher	THIS IS A SECRET (confirmed by Caesar elimination and self-inverse check)
4	Steganography (LSB)	Detect by hash/size comparison, metadata and structure checks, LSB-plane and chi-square analysis; strengthen with payload encryption + key-based scattered/transform-domain embedding